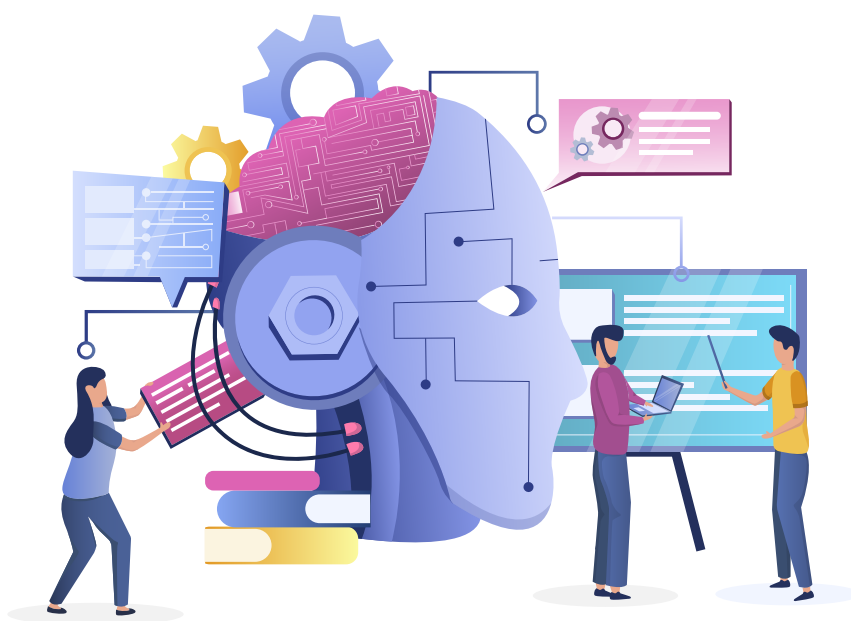


# Machine Learning: Classification Using Python

In machine learning (ML), a set of data is analysed to predict a result. Python is considered one of the best programming language choices for ML. In this article, we will discuss machine learning with respect to classification using Python.



Let's say you want to teach a child to differentiate between apples and oranges. There are various ways to do this. You could ask the child to touch both kinds of fruits so that they get familiar with the shape and the softness. You could also show her multiple examples of apples and oranges, so that they can visually spot the differences. The technological equivalent of this process is known as machine learning.

Machine learning teaches computers to solve a particular problem, and to get better at it through experience. The example discussed here is a classification problem, where the machine is given various labelled examples, and is expected to label an unlabelled sample using the knowledge it acquired from the labelled

samples. A machine learning problem can also take the form of regression, where it is expected to predict a real-valued solution to a given problem based on known samples and their solutions. Classification and regression are broadly termed as supervised learning. Machine learning can also be unsupervised, where the machine identifies patterns in unlabelled data, and forms clusters of samples with similar patterns. Another form of

machine learning is reinforcement learning, where the machine learns from its environment by making mistakes.

## Classification

Classification is the process of predicting the label of a given set of points based on the information obtained from known points. The class, or label, associated with a data set could be binary or multiple in nature. As an example, if we have

Table 1

| Age | Credit rating | Job | Property owned | Loan approved |
|-----|---------------|-----|----------------|---------------|
| 35  | good          | yes | yes            | <b>yes</b>    |
| 32  | poor          | yes | no             | <b>no</b>     |
| 22  | fair          | no  | no             | <b>no</b>     |
| 42  | good          | yes | no             | <b>yes</b>    |